

7.5 Classwork: Interest and Depreciation (calculator)

1. [Maximum mark: 9]

Hannah deposits \$2000 in a savings account that pays interest of 4% per annum, applied at the end of each year. No further deposits or withdrawals are made.

- (a) Show that the value of the account at the end of the first year is \$2080. [1]

- (b) Find the value of the account at the end of 5 years. Give your answer correct to the nearest dollar. [3]

- (c) Find the value of the account at the end of 10 years. [2]

- (d) Find the smallest number of full years needed for the value of the account to first exceed \$3000. [3]

2. [Maximum mark: 10]

Mr. Rossi buys a new car for \$30 000. Each year the value of the car decreases by 15 % of its value at the start of that year.

(a) Show that the value of the car at the end of the first year is \$25 500. [1]

(b) Find the value of the car at the end of 4 years. Give your answer correct to the nearest dollar. [3]

(c) Find the smallest number of full years after which the car is worth less than \$10 000. [3]

(d) Mr. Rossi will sell the car when it is worth less than half its original value. Find the number of years after which this first occurs. [3]